# Neural Network
# Back Propagation Training Algorithm

## CSAI 253 - Spring 2024

*Prepared By*

**Dr. Khaled Mostafa ElSayed**

kmelsayed@zewailcity.edu.eg

**Instructors:**

***Dr. Khaled Mostafa ElSayed***

kmelsayed@zewailcity.edu.eg

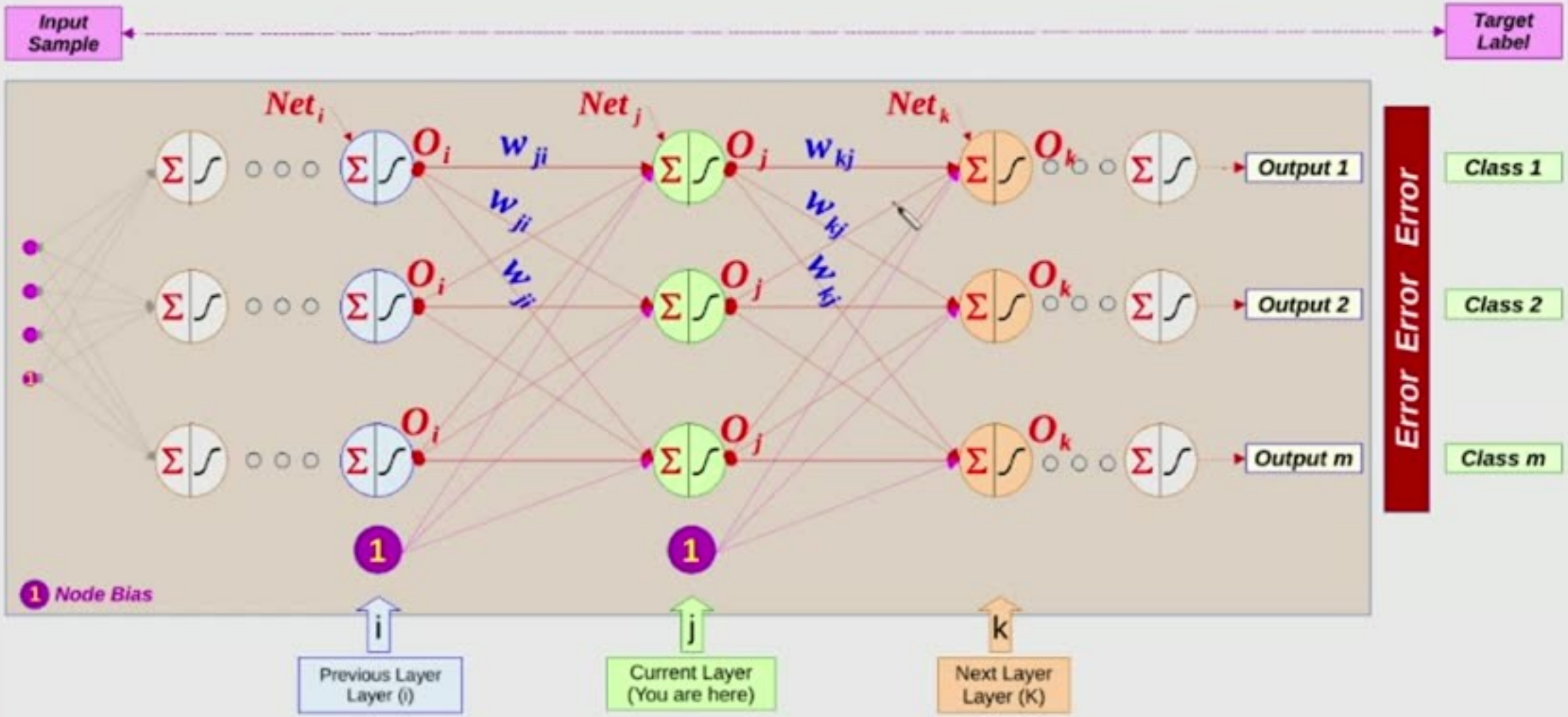
***Dr. Mohamed Maher Ata***

momaher@zewailcity.edu.eg



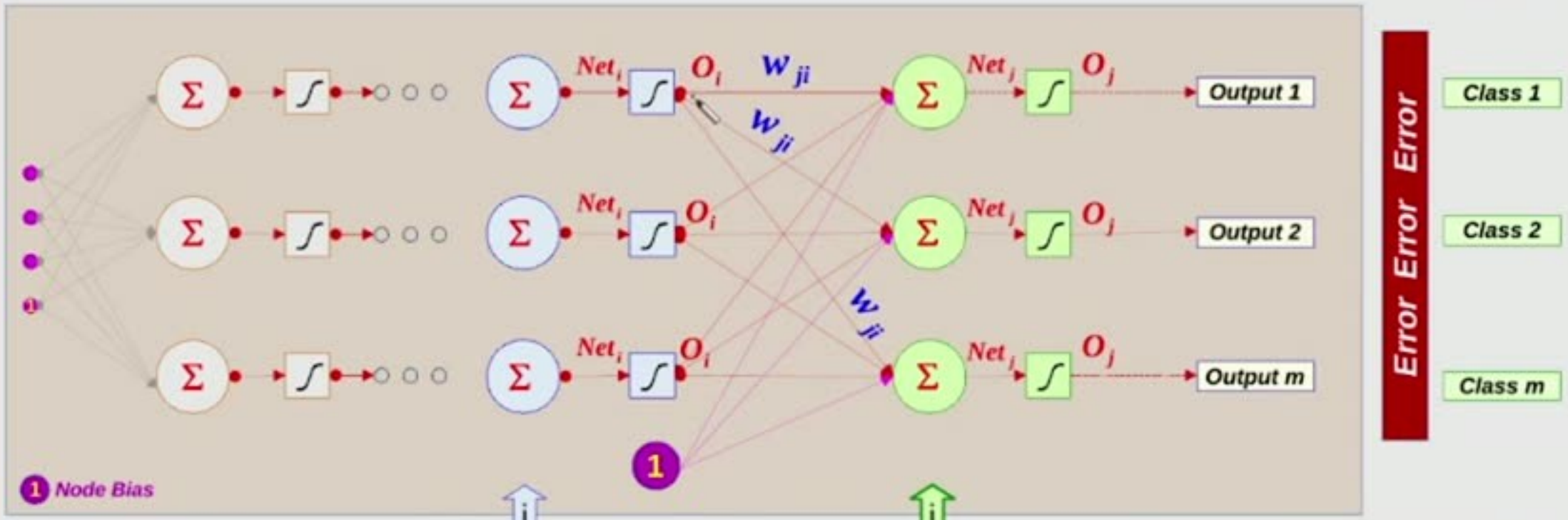
# Neural Network Model
## (Each node consists of Summation + Activation)





# Neural Network Model
## (Each node consists of Summation + Activation)






For Layer "j"

$$Net_j = \sum_i (w_{ji} O_i)$$

$$O_j = f(Net_j) = \frac{1}{1+e^{-Net_j}}$$

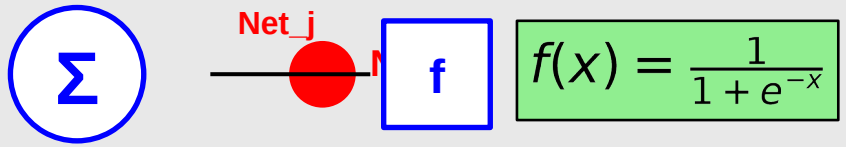
If Layer "j" is Output Layer:

$$E = \frac{1}{2} \sum_j (t_j - O_j)^2$$

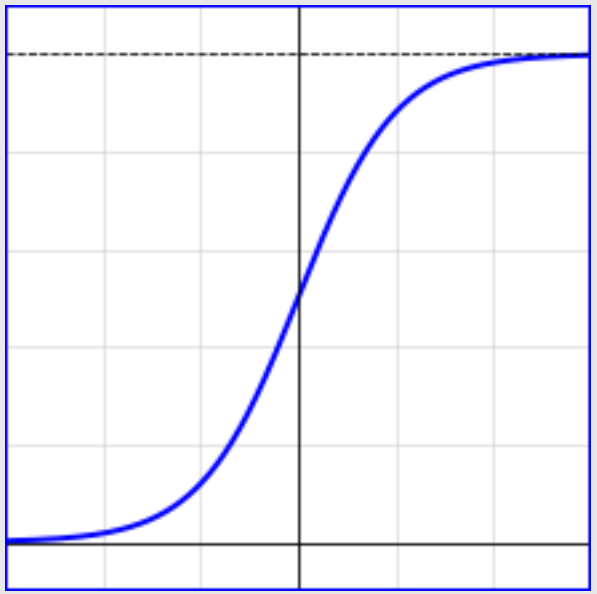
# Remember: Derivative of Sigmoid Function



$$O_j = f(Net_j) = \frac{1}{1+e^{-Net_j}}$$




$$f(x) = \frac{1}{1+e^{-x}}$$



$$f'(Net_j) = \frac{e^{-Net_j}}{(1+e^{-Net_j})^2}$$

$$f'(Net_j) = \frac{1}{1+e^{-Net_j}} \cdot \frac{(1+e^{-Net_j})-1}{1+e^{-Net_j}}$$

$$f'(Net_j) = \frac{1}{1+e^{-Net_j}}\left(1 - \frac{1}{1+e^{-Net_j}}\right)$$

$$f'(Net_j) = O_j(1-O_j)$$

# Revision



$b = f(a)\ \ c = f(b)\ \Rightarrow\ c = f(a)$ *if a is changed , c will be chnaged accordingly)*

$$\frac{\partial c}{\partial a} = \frac{\partial c}{\partial b} * \frac{\partial b}{\partial a}$$ *Chain Rule*

$= f(\underline{W}) = w_1 . a_1 + w_2 . a_2 + w_3 . a_3 + \ldots\ldots + w_n . a_n = \sum_{i=1}^{n} w_i . a_i$

$$\frac{\partial b}{\partial w_1} = a_1 \quad \frac{\partial b}{\partial w_2} = a_2 \quad \frac{\partial b}{\partial w_i} = a_i$$

**Conclusion:** $b = f(\underline{W}) = \sum_{i=1}^{n} a_i w_i$ then $\frac{\partial b}{\partial w_i} = a_i$ *One Term ONLY*

$c = f(a_1), c = f(a_2), c = f(a_3)$ and $a_1 = f(b), a_2 = f(b), a_3 = f(b)$

$c = f(b) \Rightarrow$ if b is changed, c will be changed accordingly *[through* $a_1, a_2, a_3]$

$$\frac{\partial c}{\partial b} = \left[\frac{\partial c}{\partial a_1} * \frac{\partial a_1}{\partial b}\right] + \left[\frac{\partial c}{\partial a_2} * \frac{\partial a_2}{\partial b}\right] + \left[\frac{\partial c}{\partial a_3} * \frac{\partial a_3}{\partial b}\right] = \sum_{i=1}^{3} \frac{\partial c}{\partial a_i} * \frac{\partial a_i}{\partial b}$$

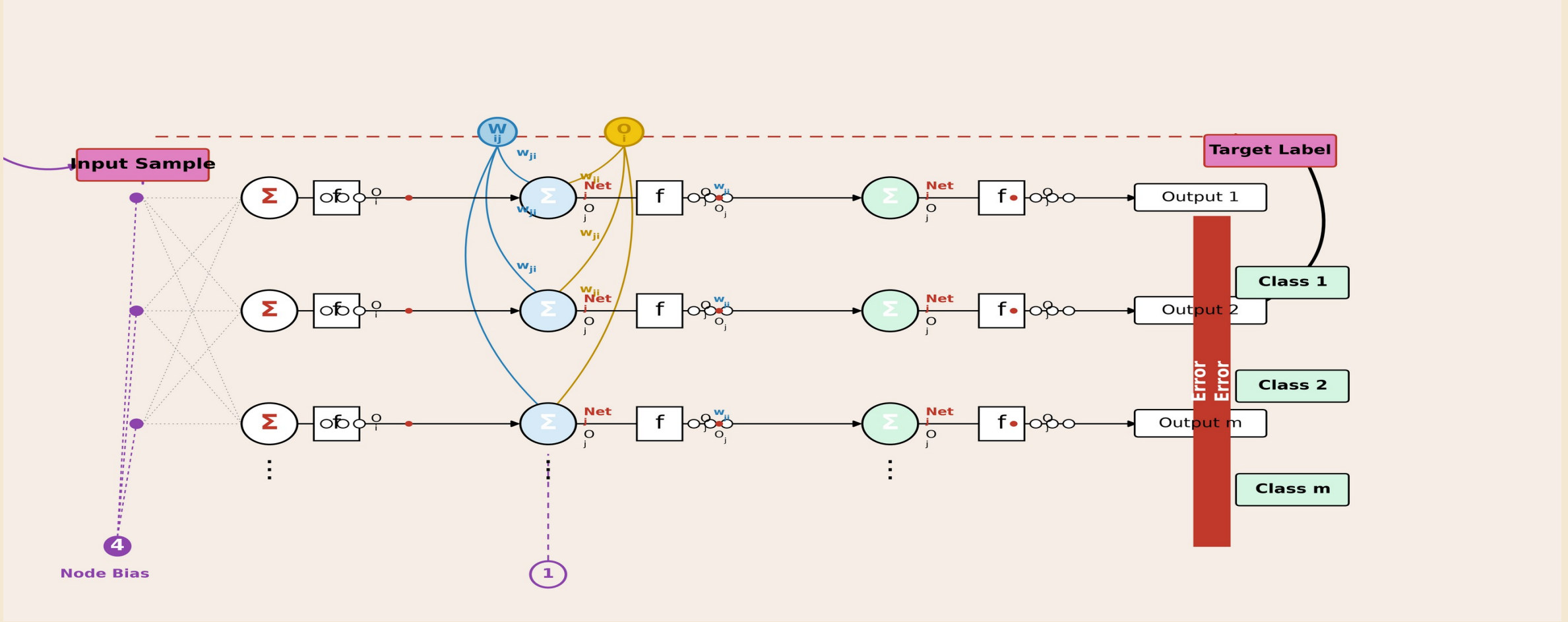
# Update Weights of Output Layer



***( Weights from previous layer to output layer)***

# Neural Network Model

## (Weights of Output Layer AFFECTS Calculated Error)




For Layer "j"

$$Net_j = \sum w_{ji} O_i$$

$$O_j = f(Net_j) = \frac{1}{1 + e^{-Net_j}}$$

↑

**Previous Layer Layer (0)**

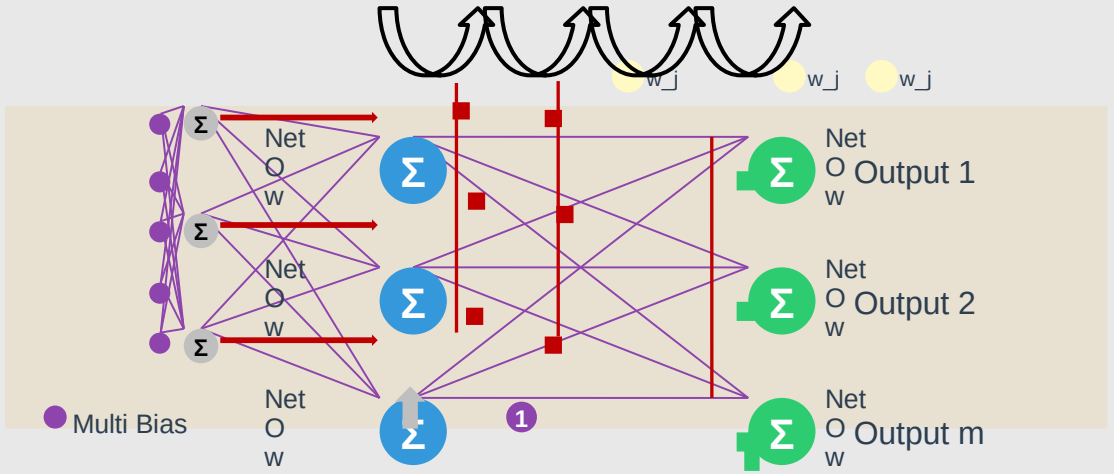
↑

**Current Layer is Output Layer (You are here)**

If Layer "j" is Output Layer:

$$E = \frac{1}{2} \sum (t_j - O_j)^2$$

$\frac{\partial E}{\partial w_{ij}}$




?

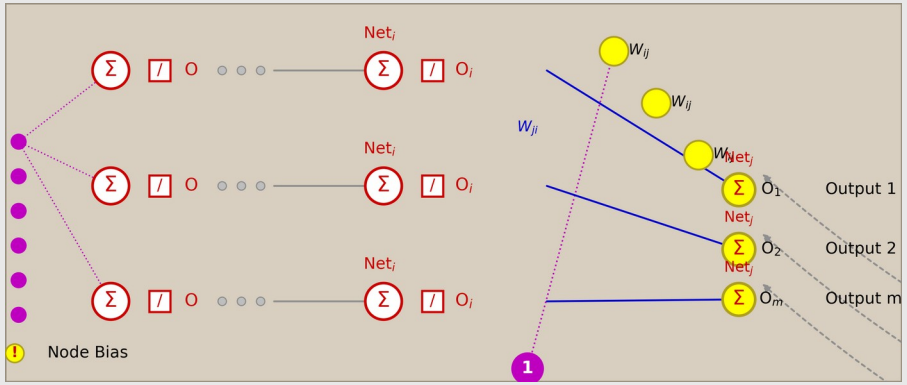
# Gradient Descent



$$W_{ji}^{Updated} = W_{ji}^{old} - \eta \frac{\partial E}{\partial W_{ji}}$$

$$W_{ji}^{Updated} = W_{ji}^{old} + \Delta_{ji}$$

$$\text{Where } \Delta_{ji} = -\eta \frac{\partial E}{\partial W_{ji}}$$




Error

Error Class 1

Error Class 2

Error Class m

$$E = \frac{1}{2}\sum_j (t_j - O_j)^2$$

*For Sigmoid Activation* $O_j = f(Net_j) = \frac{1}{1+e^{-Net_j}}$

$$Net_j = \sum_i (w_{ji} O_i)$$

$$\frac{\partial E}{\partial O_j} = -(t_j - O_j)$$

$$\frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

$$\frac{\partial Net_j}{\partial W_{ji}} = O_i$$

$$\frac{\partial E}{\partial W_{ji}} = \frac{\partial E}{\partial O_j} * \frac{\partial O_j}{\partial Net_j} * \frac{\partial Net_j}{\partial W_{ji}}$$

$$= -(t_j - O_j) * O_j(1 - O_j) * O_i$$

$$\frac{\partial E}{\partial O_j} = -(t_j - O_j)$$

$$\frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

$$\frac{\partial Net_j}{\partial W_{ji}} = O_i$$

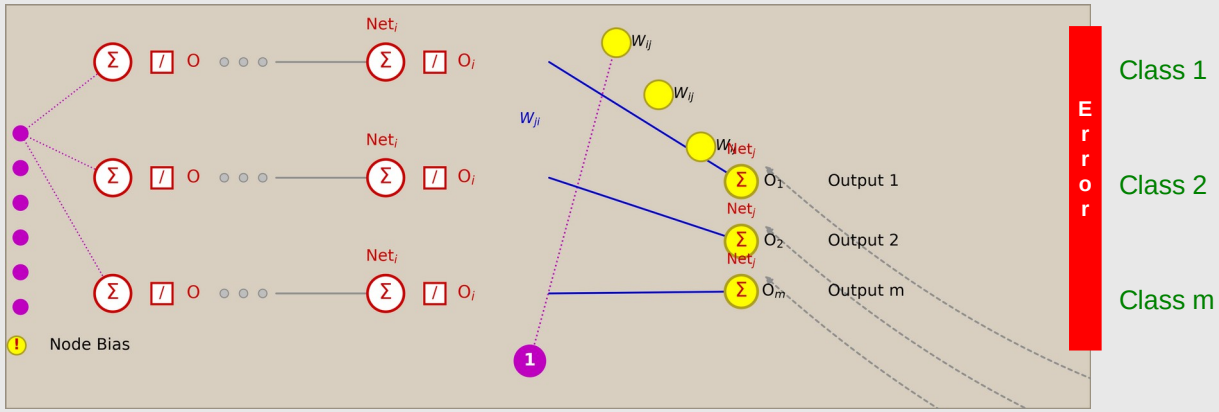
# Gradient Descent



$$W_{ji}^{updated} = W_{ji}^{old} - \eta \frac{\partial E}{\partial W_{ji}}$$

$$W_{ji}^{updated} = W_{ji}^{old} + \Delta_{ji}$$

**Where** $\Delta_{ji} = -\eta \frac{\partial E}{\partial W_{ji}}$




$$E = \frac{1}{2}\sum_j (t_j - O_j)^2$$

***For Sigmoid Activation***

$$O_j = f(Net_j) = \frac{1}{1 + e^{-Net_j}}$$

$$Net_j = \sum_i (w_{ji} O_i)$$

$$\frac{\partial E}{\partial O_j} = -(t_j - O_j)$$

$$\frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

$$\frac{\partial Net_j}{\partial W_{ji}} = O_i$$

$$\frac{\partial E}{\partial W_{ji}} = \frac{\partial E}{\partial O_j} \cdot \frac{\partial O_j}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial W_{ji}} = \frac{\partial E}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial W_{ji}}$$

$$= -(t_j - O_j) \cdot O_j(1 - O_j) \cdot O_i = -\delta_j \cdot O_i$$

**Where** $\delta_j = -\frac{\partial E}{\partial Net_j} = (t_j - O_j) \cdot O_j(1 - O_j)$ ($\delta_j$ depends on $j$ ONLY)

$$\frac{\partial E}{\partial O_j} = -(t_j - O_j)$$

$$\frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

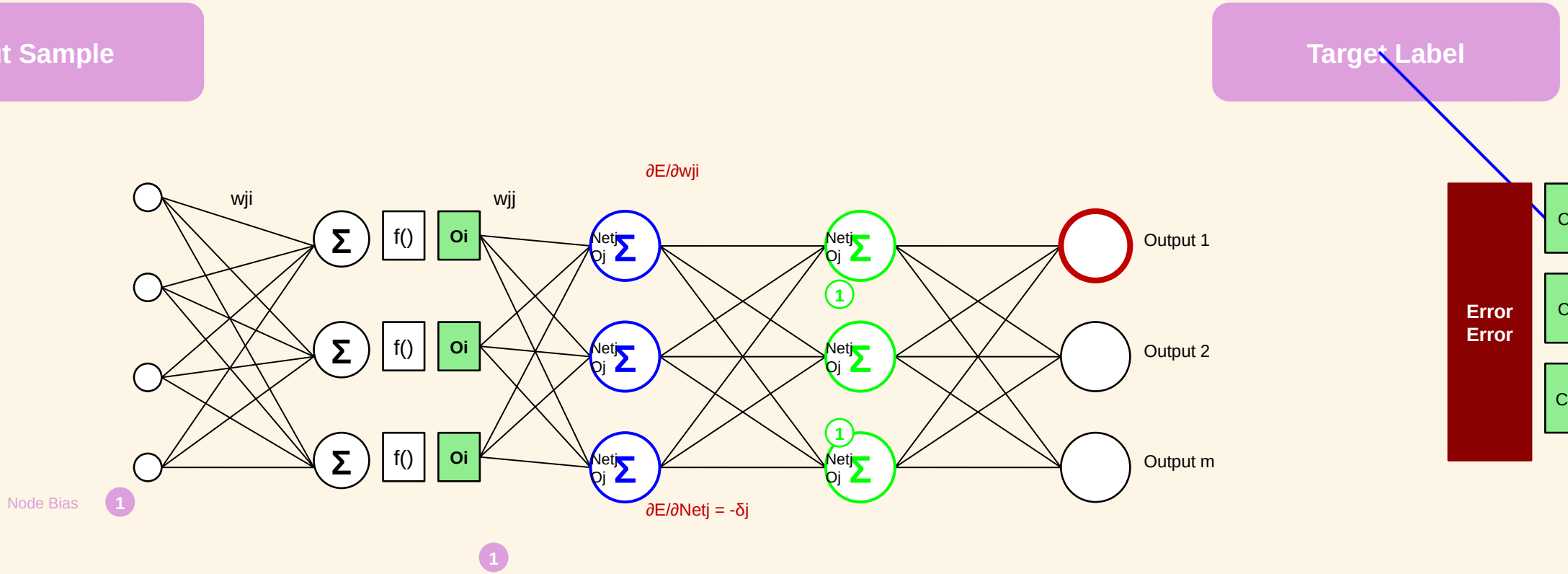
$$\frac{\partial Net_j}{\partial W_{ji}} = O_i$$

$$\Delta_{ji} = -\eta \frac{\partial E}{\partial W_{ji}} = \eta \delta_j \cdot O_i$$ **Where** $\delta_j = (t_j - O_j) \cdot O_j(1 - O_j)$ for output Layer

# Updated Weights of Output Layer



( Calculated Error is affected by Netj,)



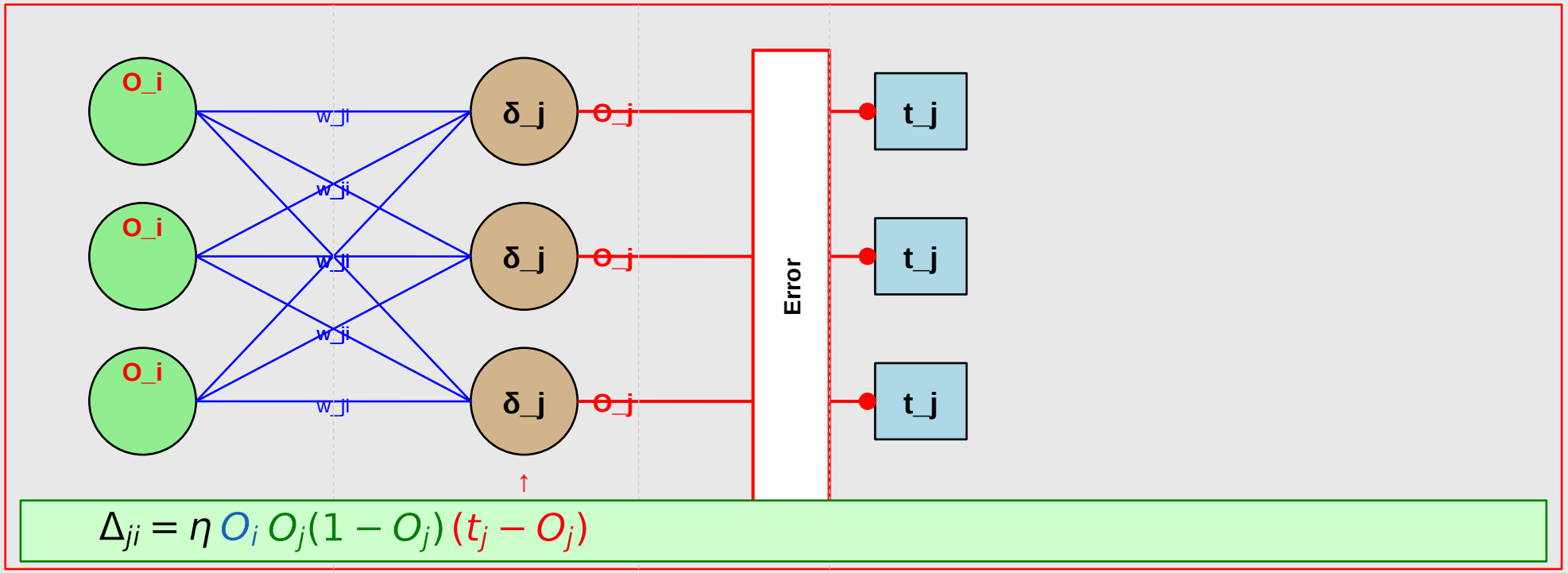

$$\Delta_{ji} = -\eta \frac{\partial E}{\partial w_{ji}} = -\frac{\partial E}{\partial Net_j}\frac{\partial Net_j}{\partial w_{ji}} = \eta \delta_j \cdot O_i$$

Where δj = -∂E/∂Netj (one δ for each Node j)

Layer, Sigmoid Activation $\Rightarrow \delta_j = (t_j - O_j) \cdot O_j(1 - O_j) \Rightarrow \Delta_{ji} = \eta(t_j - O_j) \cdot O_j(1 - O_j) \cdot O_i$

# At Output Layer






$\Delta_{ji} = \eta\, O_i\, \delta_j$

Where: $\delta_j = O_j(1 - O_j)\,(t_j - O_j)$

# Update Weights of Hidden Layer

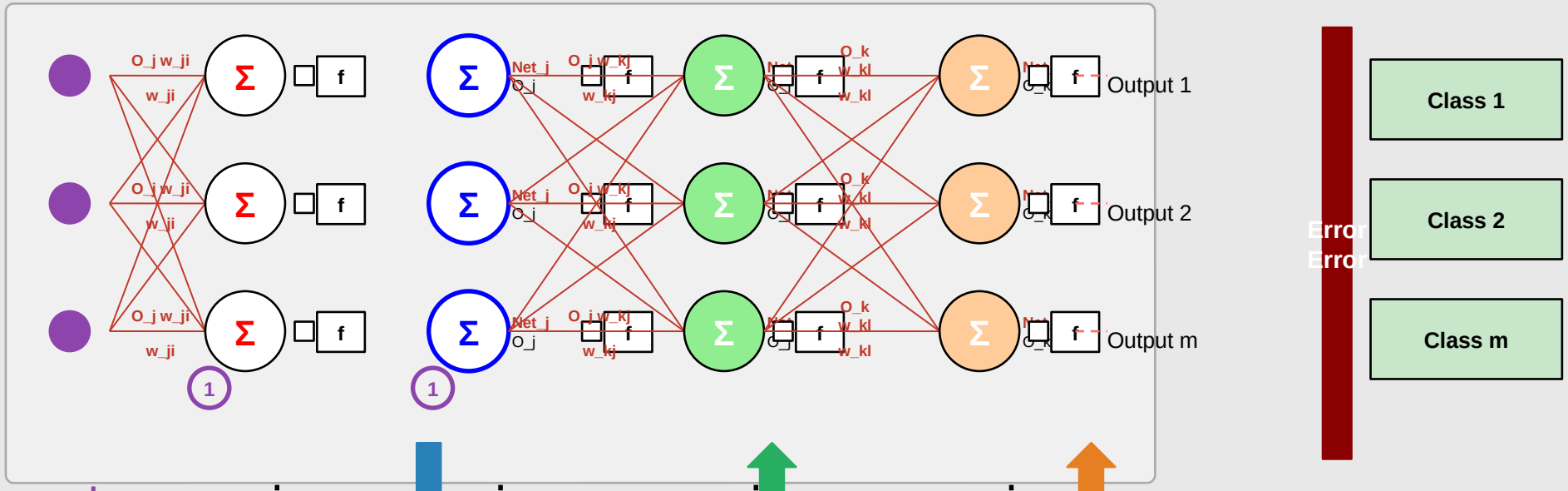
*(Weights from previous layer to Hidden layer)*

# Neural Network Model

## (Current Layer is Hidden Layer)

Input Sample

Target Label




For Layer "j"

$$Net_j = \Sigma_i(w_{ji} \cdot O_i)$$

$$O_j = f(Net_j) = \frac{1}{1 + e^{-Net_j}}$$

Previous Layer
(I)

Current Layer
(You are here)

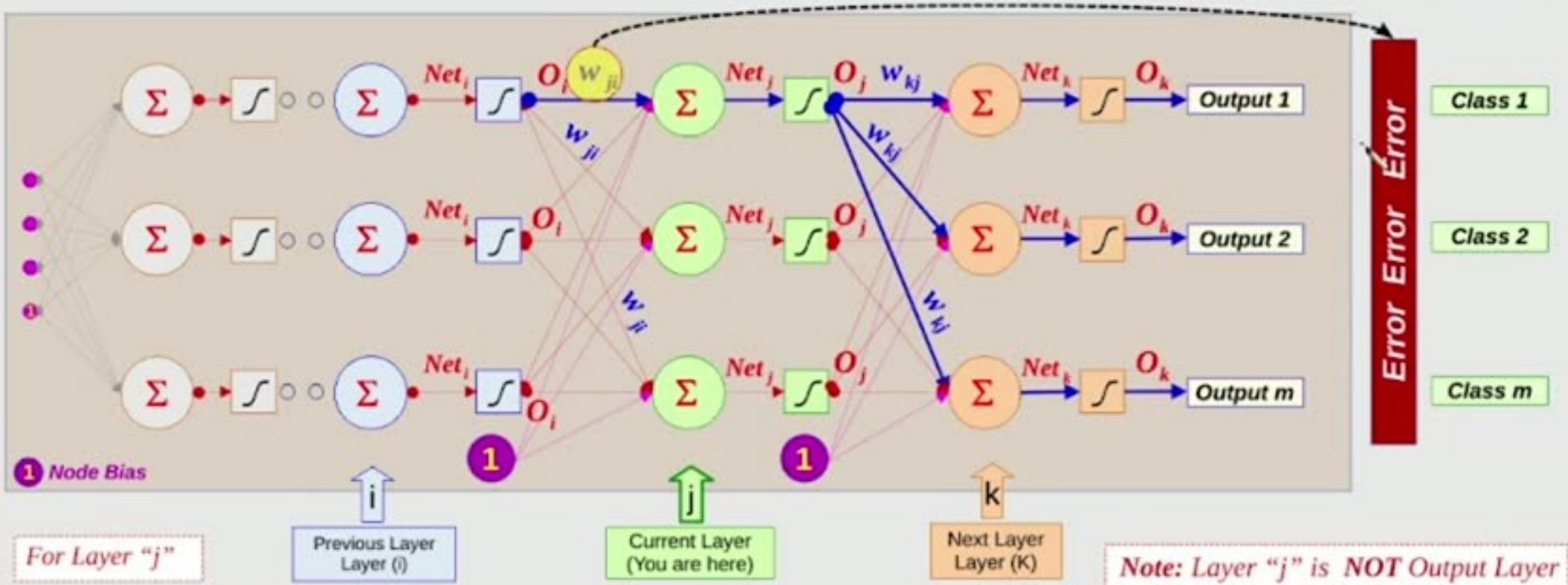
Next Layer
(K)

Note: Layer "j" is **NOT** Output Layer

$$E = \frac{1}{2}\Sigma_j(t_j - O_j)^2$$

X X X

# Neural Network Model
## (Weights of hidden layer AFFECT Calculated Error)




For Layer "j"

$$Net_j = \sum_i (w_{ji} O_i)$$

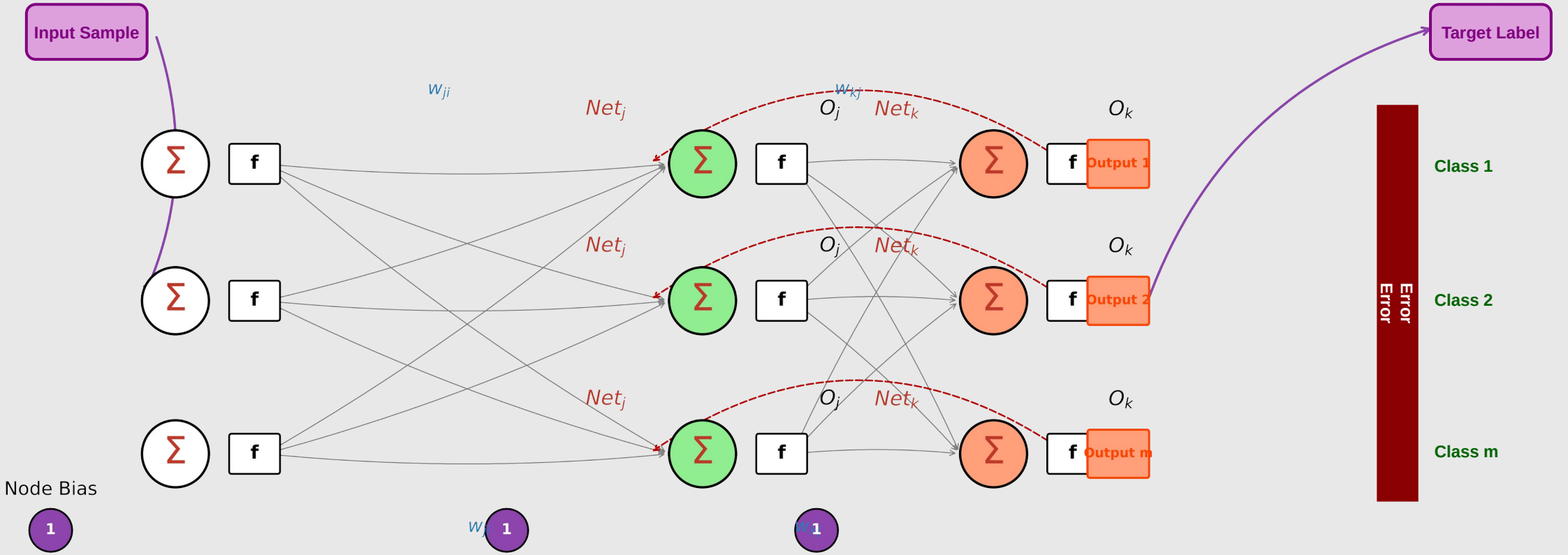
$$O_j = f(Net_j) = \frac{1}{1+e^{-Net_j}}$$

**Note:** Layer "j" is **NOT** Output Layer

$$E = \frac{1}{2} \sum_j (t_j - O_j)^2$$ XXX

# Neural Network Model

## (How Weights of hidden layer affect Calculated Error)




↑ i

For Layer "j"

Previous Layer (I)

↑ j

Current Layer (You are here)

↑ k

Next Layer (K)

Note: Layer "j" is NOT Output Layer

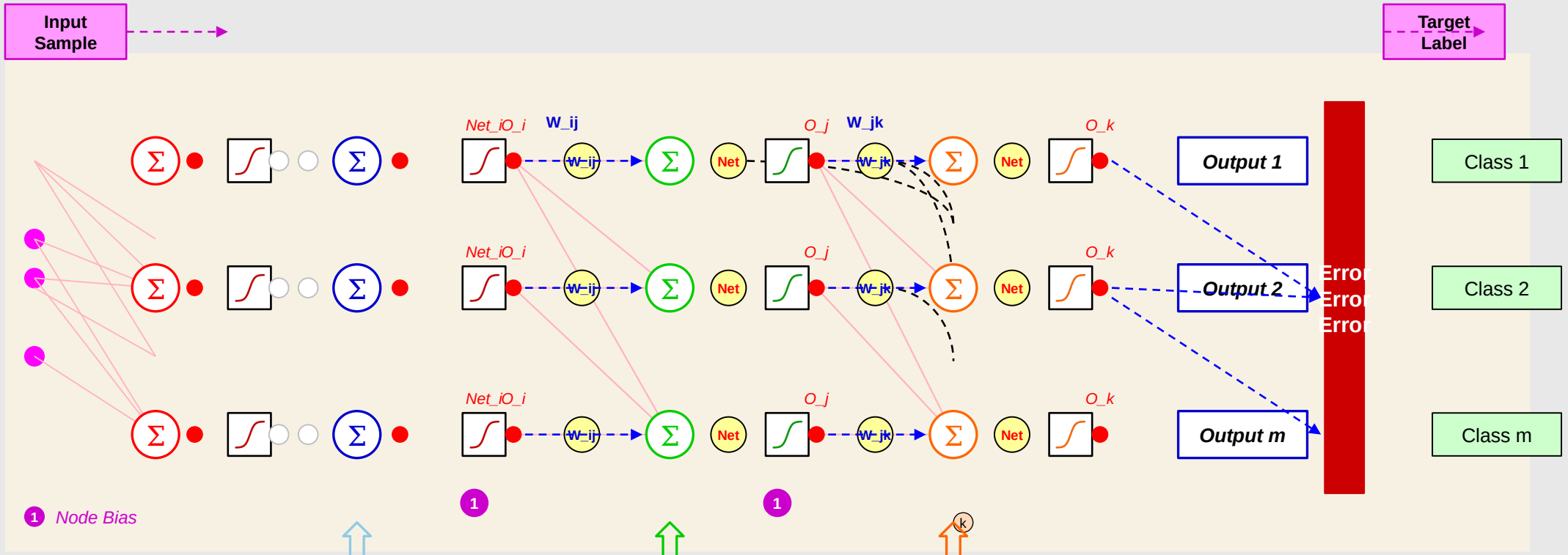
$$Net_j = \sum_i (w_{ji} O_i)$$

$$O_j = f(Net_j) = \frac{1}{1 + e^{-Net_j}}$$

$$E = \frac{1}{2} \sum_j (t_j - O_j)^2 \quad \text{X X X}$$

# Neural Network Model

**(Effect of Net_k on Calculated Error is known from O/P Layer)**




*For Layer "j"*

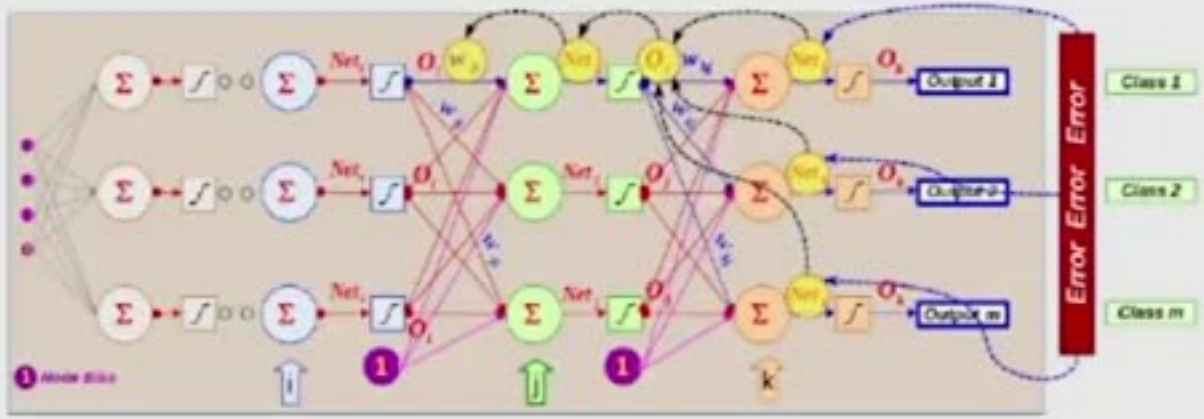
$$Net_j = \sum_i w_{ij} O_i$$

$$O_j = f(Net_j) = \frac{1}{1 + e^{-Net_j}}$$

*Note: Layer "j" is NOT Output Layer*

X X X

$$E = \frac{1}{2} \sum_j (t_j - O_j)^2$$



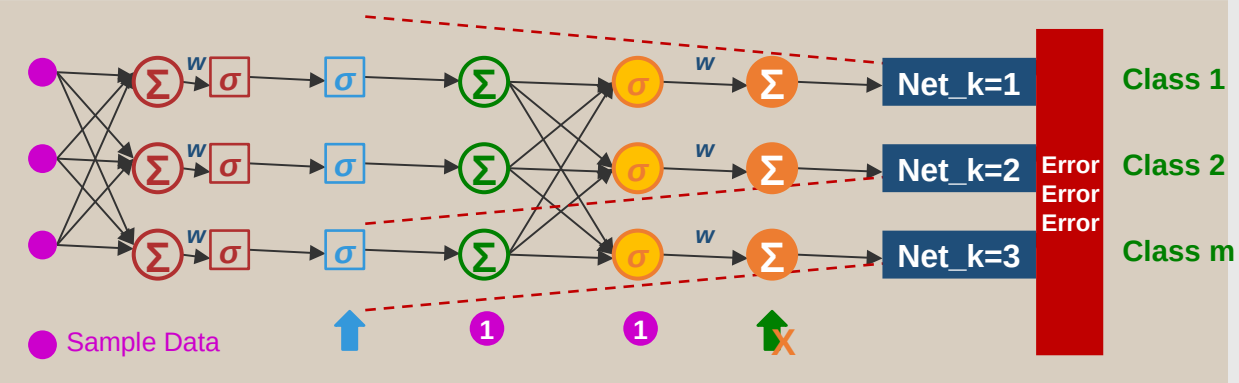

$$\frac{\partial E}{\partial w_{ji}} ?$$

$W_{ji}$ *Affects* $Net_j$ ⇒

$Net_j$ *Affects* $O_j$ ⇒

$O_j$ *Affects* $(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ ⇒

$(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ *Affect Error*




$$= \left[ \frac{\partial E}{\partial \text{Net}_{k=1}} \cdot \frac{\partial \text{Net}_{k=1}}{\partial O_j} + \frac{\partial E}{\partial \text{Net}_{k=2}} \cdot \frac{\partial \text{Net}_{k=2}}{\partial O_j} + \cdots + \frac{\partial E}{\partial \text{Net}_{k=m}} \cdot \frac{\partial \text{Net}_{k=m}}{\partial O_j} \right] \cdot \frac{\partial O_j}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial w_{ji}}$$

$$\frac{\partial E}{\partial Net_k} \Rightarrow -\delta_k$$

$$\text{Net}_k = \sum_k w_{kj} O_j \Rightarrow \frac{\partial \text{Net}_k}{\partial O_j} = w_{kj}$$

$$O_j = \frac{1}{1 + e^{-Net_j}} \Rightarrow \frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

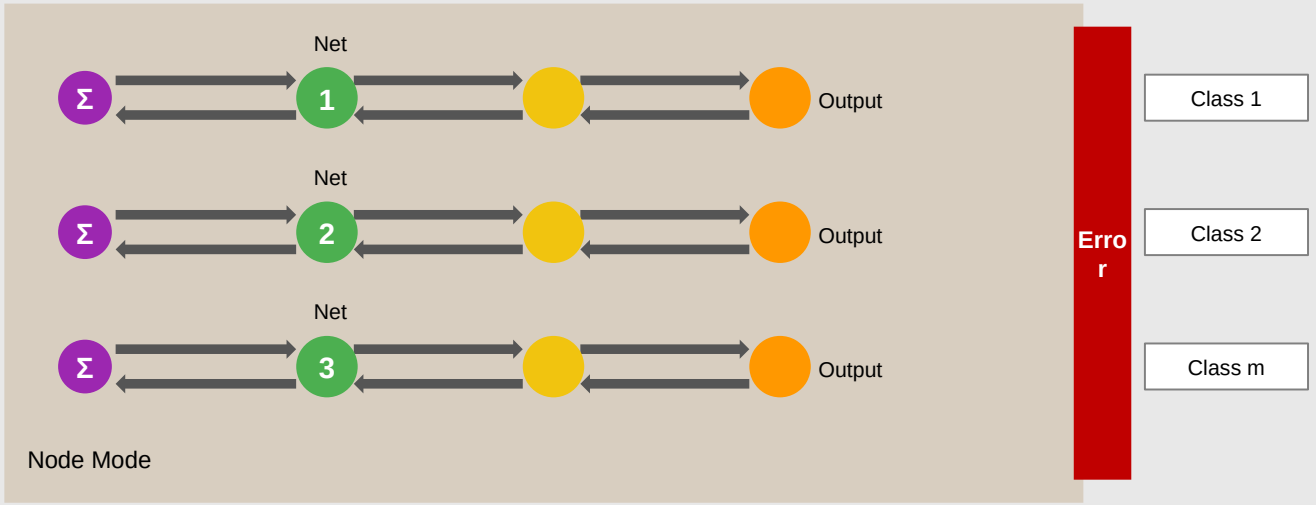
$$\text{Net}_j = \sum_j w_{ji} O_i \Rightarrow \frac{\partial \text{Net}_j}{\partial w_{ji}} = O_i$$

$W_{ji}$ *Affects* $Net_j$ $\Rightarrow$

$Net_j$ *Affects* $O_j$ $\Rightarrow$

$O_j$ *Affects* $(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ $\Rightarrow$

$(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ *Affect Error*




$$\frac{\partial E}{\partial w_{ji}} = \frac{\partial E}{\partial Net_{k_1}}\frac{\partial Net_{k_1}}{\partial O_j} + \frac{\partial E}{\partial Net_{k_2}}\frac{\partial Net_{k_2}}{\partial O_j} + \ldots + \frac{\partial E}{\partial Net_{k_m}}\frac{\partial Net_{k_m}}{\partial O_j} \cdot \frac{\partial O_j}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial w_{ji}}$$

$$\frac{\partial E}{\partial w_{ji}} = \sum_{k=1}^{m} \frac{\partial E}{\partial Net_k}\frac{\partial Net_k}{\partial O_j} \cdot \frac{\partial O_j}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial w_{ji}} = \sum_{k=1}^{m} -\delta_k w_{kj} \cdot O_j(1-O_j) \cdot O_i$$

$$\Delta_{ji} = -\eta\frac{\partial E}{\partial w_{ji}} = \sum_{k=1}^{m} \delta_k w_{kj} \cdot O_j(1-O_j) \cdot O_i = \eta\delta_j \cdot O_i$$

Where $\delta_j = \sum_{k=1}^{m} \delta_k w_{kj} \cdot O_j(1-O_j)$ (for each Node $j$ at hidden layer)

$$\frac{\partial E}{\partial Net_k} \rightarrow -\delta_k$$

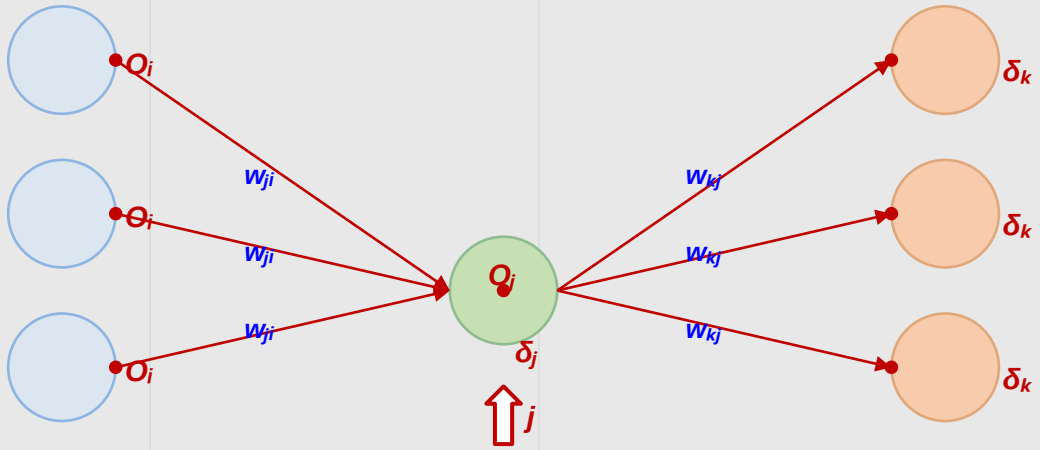
$$Net_k = \sum_j w_{kj} O_j \rightarrow \frac{\partial Net_k}{\partial O_j} = w_{kj}$$

$$O_j = \frac{1}{1+e^{-Net_j}} \rightarrow \frac{\partial O_j}{\partial Net_j} = O_j(1-O_j)$$

$$Net_j = \sum_i w_{ji} O_i \rightarrow \frac{\partial Net_j}{\partial w_{ji}} = O_i$$

# At Hidden Layer



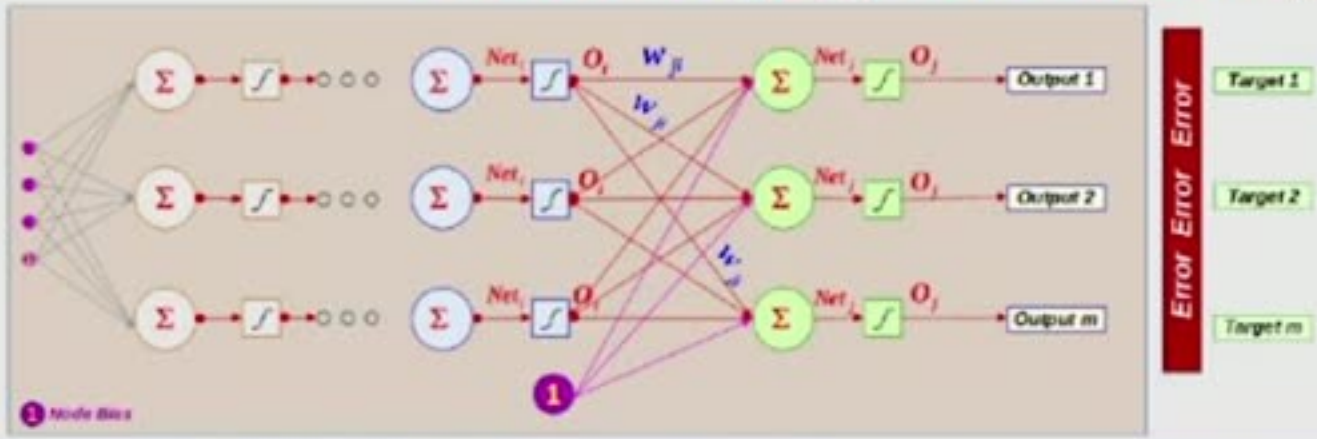



$$\Delta_{ji} = \eta O_i O_j (1 - O_j)[\sum_k (\delta_k w_{kj})]$$

$$\Delta_{ji} = \eta O_i \delta_j$$

Where : $\delta_j = O_j (1 - O_j)[\sum_k (\delta_k w_{kj})]$

# Back Propagation Training Algorithm




**[1]** Initialize all Weights with random values
**[2]** Provide one training sample at a time (and its corresponding target label)

**[3]** *Go Forward* : Start from Input Layer, Calculate $Net_j, O_j$ for each layer till reach output layer

**[4]** *Go Backward* : Start from Output Layer, Calculate $\Delta_{ji}, W_{ji}$ for each weights of ***Output*** layer

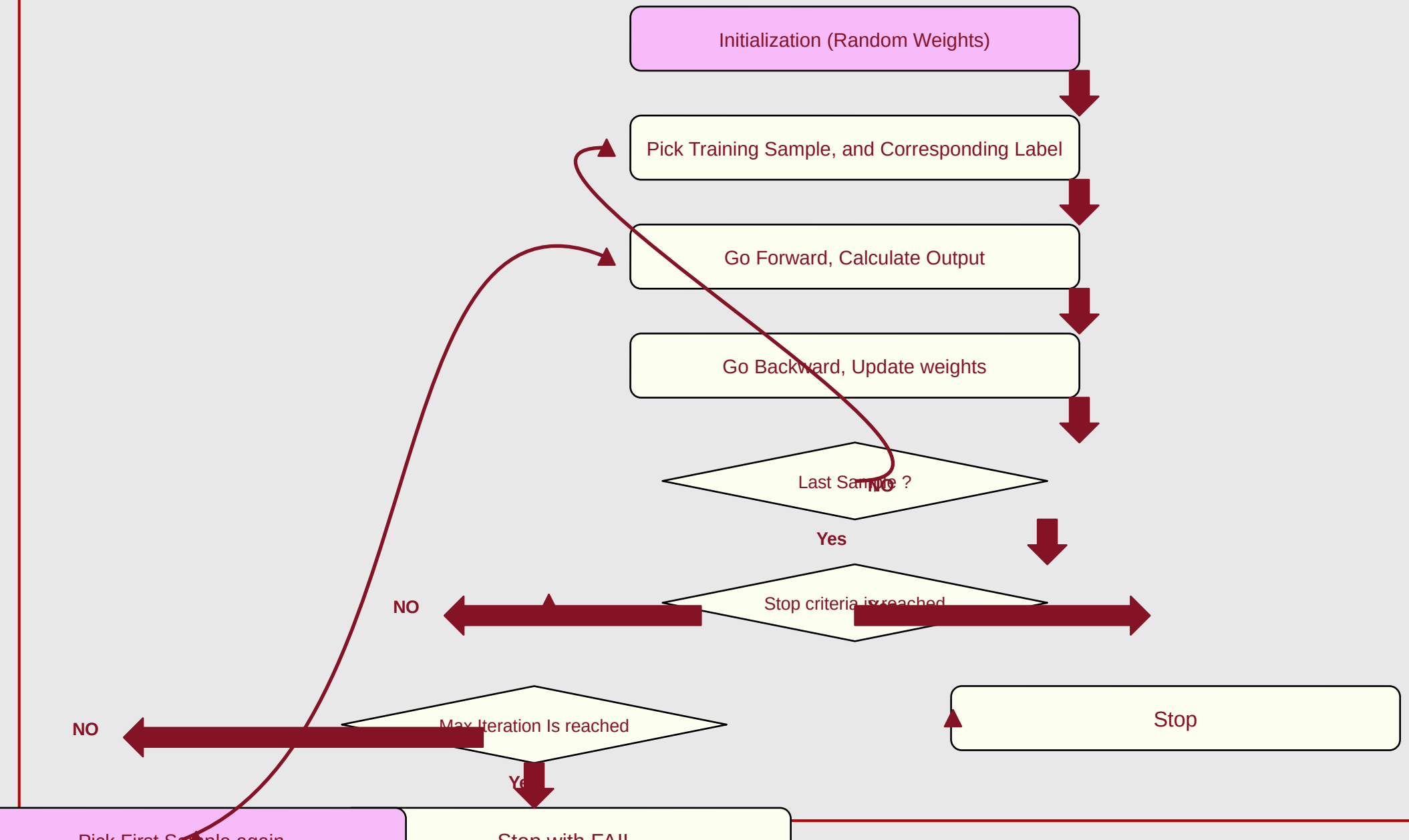
$\Delta_{ji} = \eta \; O_i \; O_j(1-O_j) \; (t_j - O_j)$ **Store:** $\delta_j = O_j(1-O_j) \; (t_j - O_j)$ *(for each node at output Layer)*

**[5]** *Go Backward* : go to previous Layer, Calculate $\Delta_{ji}, W_{ji}$ for each weights of ***Hidden*** layer

$\Delta_{ji} = \eta \; O_i \; O_j(1-O_j) \; [\sum_k (\delta_k w_{kj})]$ **Store:** $\delta_j = O_j(1-O_j) \; [\sum_k (\delta_k w_{kj})]$ *(for each node at hidden Layer)*

**[6]** *Repeat* **[5]** *till reach first hidden layer*

Remember: $\delta_k = \delta_j$ of next layer ( which is already calulated in previous step )


Initialization (Random Weights)
Pick Training Sample, and Corresponding Label
Go Forward, Calculate Output
Go Backward, Update weights
Last Sample ?
NO
Yes
NO
Stop criteria is reached
NO
Max Iteration Is reached
Stop

# Stopping Criteria



[1] Total Error = zero (Theoretical)

[2] Total Error < certain value (Threshold)

[3] Percentage Change of Weights is less than certain value (Threshold)

[4] Reach Maximum number of iterations (Epochs) without satisfying [2] or [3]

**Stopping is due to reaching Maximum number of iterations = Training is FAILED**

Expected Reasons:

[1] The defined Maximum Number of Epochs is small. Try to increase the limit

[2] Under-fitting (Training samples are not adequate). Try to provide more training samples

[3] Model Design is not suitable to training data. Try to add more nodes in hidden layer or add more hidden Layers

[4] Data is NOT separable by Neural Network. Try Another Models (e.g. SVM with Kernels)

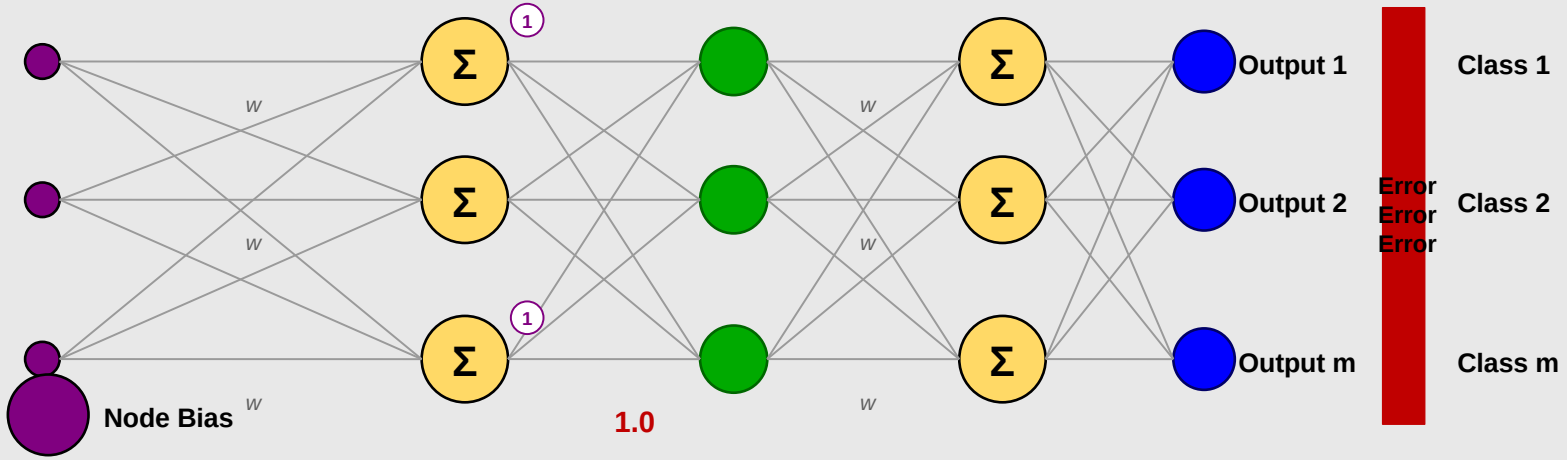
# Vanishing and Exploding Gradients



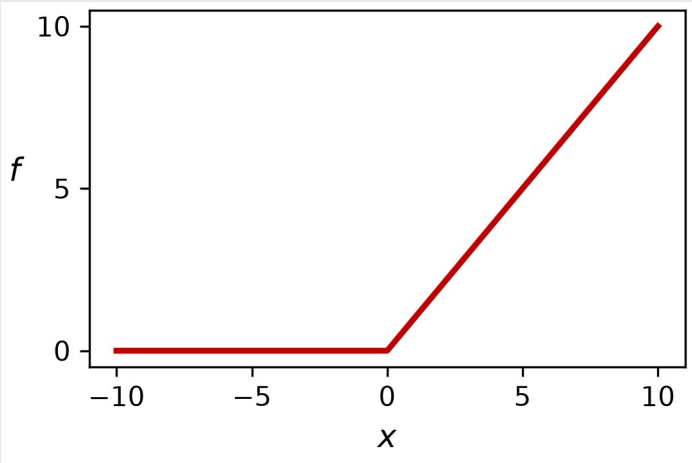
## Long Chain of Derivatives

$$f(x) = \text{ReLU} = \begin{cases} 0 \text{ for } x < 0 \\ x \text{ for } x > 0 \end{cases}$$

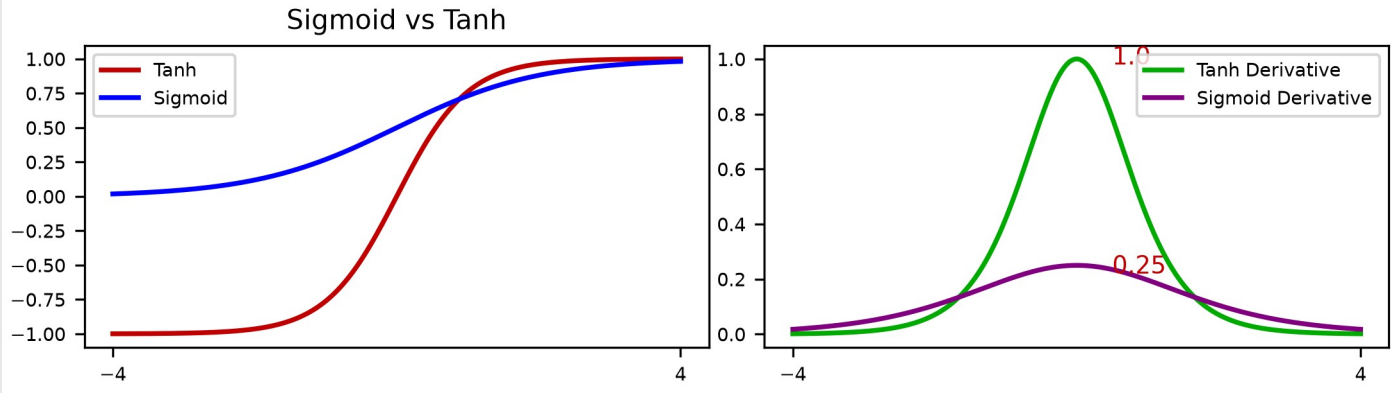
$$f'(x) = \begin{cases} 0 \text{ for } x < 0 \\ 1 \text{ for } x \geq 0 \end{cases}$$




## Exploding Gradients




## Vanishing Gradients

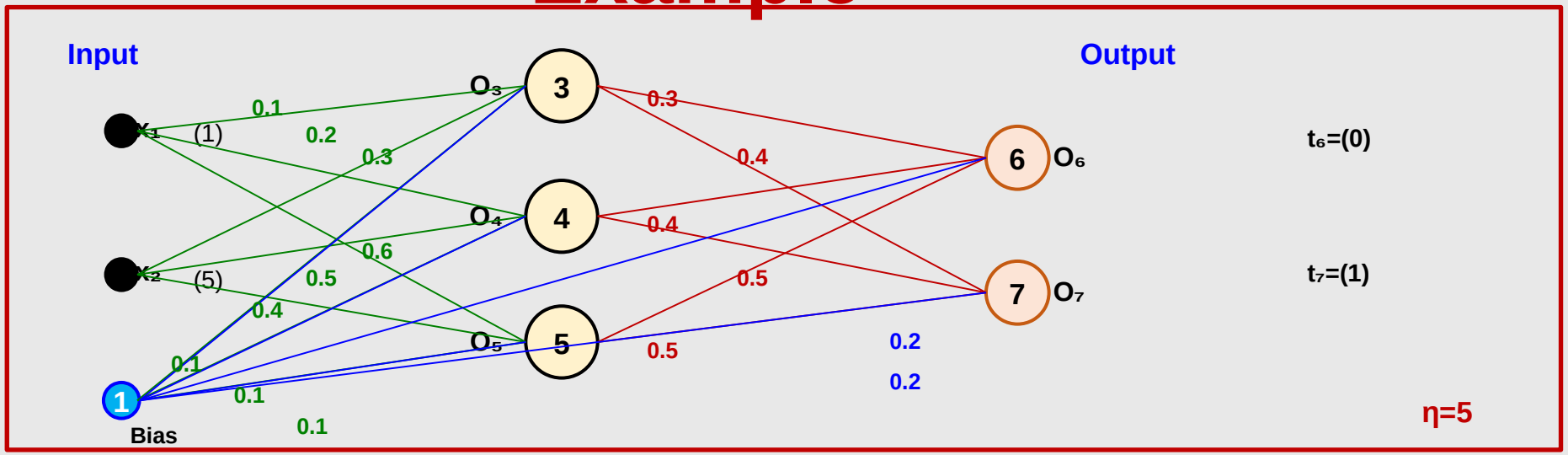


# Solved Example

[ Update Weights of Output Layer and Hidden Layer ]

Using one training sample

# Example




## Forward

$Net_3 = 1*0.1+5*0.4+1*0.1 = 2.2$

$Net_4 = 1*0.2+5*0.5+1*0.1 = 2.8$

$Net_5 = 1*0.3+5*0.6+1*0.1 = 3.4$

$$O_3 = \frac{1}{1+e^{-2.2}} = 0.9$$

$$O_4 = \frac{1}{1+e^{-2.8}} = 0.943$$

$$O_5 = \frac{1}{1+e^{-3.4}} = 0.968$$

$Net_6 = 0.9*0.3+0.943*0.4+0.968*0.5+1*0.2 = 1.3312$

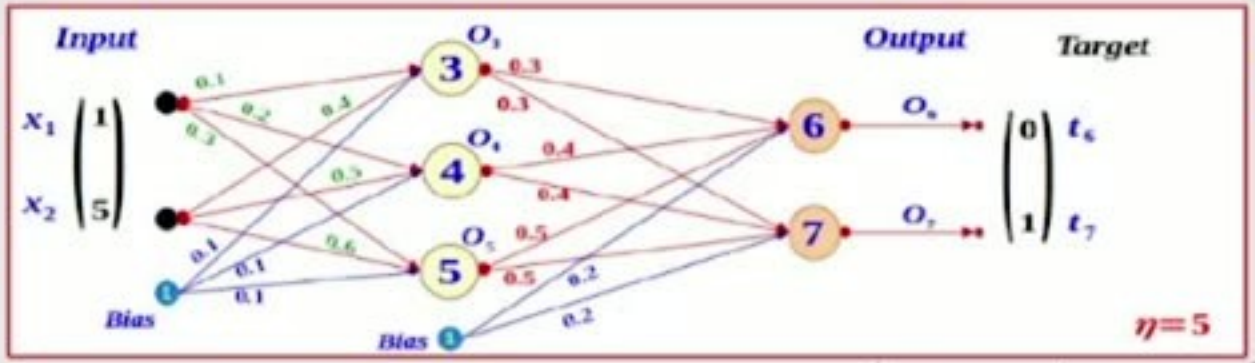
$Net_7 = 0.9*0.3+0.943*0.4+0.968*0.5+1*0.2 = 1.3312$

$$O_6 = \frac{1}{1+e^{-1.3312}} = 0.791$$

$$O_7 = \frac{1}{1+e^{-1.3312}} = 0.791$$

$$\delta_j = O_j(1-O_j)\,(t_j-O_j)$$

$$\Delta_{ji} = \eta\ O_i\ O_j(1-O_j)\ (t_j-O_j) = \eta\ O_i\ \delta_j$$




## Backward ( Update Output Layer Weights)

$$\delta_6 = 0.791(1-0.791)(0-0.791) = -0.131$$
$$\delta_7 = 0.791(1-0.791)(1-0.791) = 0.035$$

$$O_3 = \frac{1}{1+e^{-2.2}} = 0.9$$
$$O_4 = \frac{1}{1+e^{-2.8}} = 0.943$$
$$O_5 = \frac{1}{1+e^{-3.4}} = 0.968$$

$$\Delta W_{63} = 5*(-0.131)*0.9 = -0.59$$
$$\Delta W_{64} = 5*(-0.131)*0.943 = -0.618$$
$$\Delta W_{65} = 5*(-0.131)*0.968 = -0.634$$
$$\Delta W_{6b} = 5*(-0.131)*1.0 = -0.655$$

$$W_{63} = 0.3-0.59 = -0.29$$
$$W_{64} = 0.4-0.18 = -0.218$$
$$W_{65} = 0.5-0.634 = -0.134$$
$$W_{6b} = 0.2-0.655 = -0.455$$

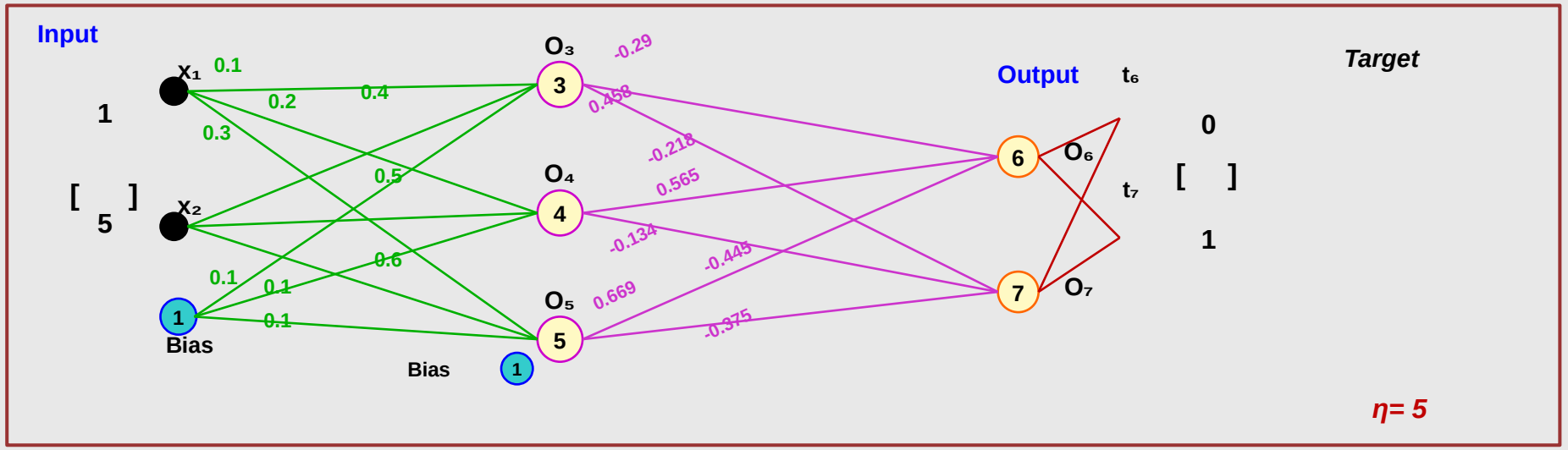
$$O_6 = \frac{1}{1+e^{-1.3312}} = 0.791$$
$$O_7 = \frac{1}{1+e^{-1.3312}} = 0.791$$

$$\Delta W_{73} = 5*(0.035)*0.9 = 0.158$$
$$\Delta W_{74} = 5*(0.035)*0.943 = 0.165$$
$$\Delta W_{75} = 5*(0.035)*0.968 = 0.169$$
$$\Delta W_{7b} = 5*(0.035)*1.0 = 0.175$$

$$W_{73} = 0.3+0.158 = 0.458$$
$$W_{74} = 0.4+0.165 = 0.565$$
$$W_{75} = 0.5+0.169 = 0.669$$
$$W_{7b} = 0.2+0.175 = 0.375$$

# After Updating O/p Layer weights



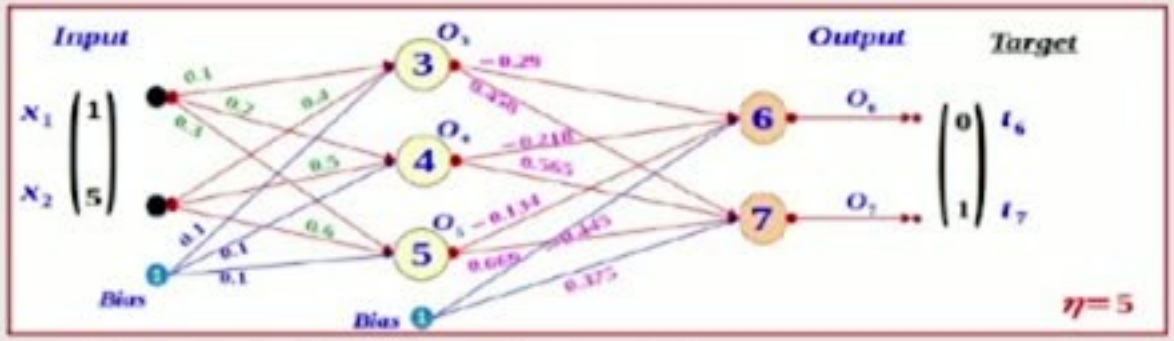


# Example



$$\delta_j = O_j(1-O_j)\ [\sum_k (\delta_k w_{kj})]$$

$$\Delta_{ji} = \eta\ O_i\ O_j(1-O_j)\ [\sum_k (\delta_k w_{kj})] = \eta\ O_i\ \delta_j$$




$\delta_6 = 0.791(1-0.791)(0-0.791) = -0.131$

$\delta_7 = 0.791(1-0.791)(1-0.791) = 0.035$

$$O_3 = \frac{1}{1+e^{-2.2}} = 0.9$$

$$O_4 = \frac{1}{1+e^{-2.8}} = 0.943$$

$$O_5 = \frac{1}{1+e^{-3.4}} = 0.968$$

## *Backward* ( Update *hidden* Layer Weights)

$\delta_3 = O_3(1-O_3)[\delta_6 * W_{63} + \delta_7 * W_{73}]$

$\delta_3 = 0.9(1-0.9)[-0.131 * -0.29 + 0.035 * 0.458] = 0.005$

$\delta_4 = 0.943(1-0.9943)[-0.131 * -0.218 + 0.035 * 0.565] = 0.003$

$\delta_5 = 0.968(1-0.968)[-0.131 * -0.134 + 0.035 * 0.669] = 0.001$

$$O_6 = \frac{1}{1+e^{-1.3312}} = 0.791$$

$$O_7 = \frac{1}{1+e^{-1.3312}} = 0.791$$

$\Delta W_{31} = 5 * 0.005 * 1 = 0.025$ → $W_{31} = 0.1 + 0.025 = 0.125$

$\Delta W_{32} = 5 * 0.005 * 5 = 0.125$ → $W_{32} = 0.4 + 0.125 = 0.525$

$\Delta W_{3b} = 5 * 0.005 * 1 = 0.025$ → $W_{3b} = 0.1 + 0.025 = 0.125$

$\Delta W_{41} = 5 * 0.003 * 1 = 0.015$ → $W_{41} = 0.2 + 0.015 = 0.215$

$\Delta W_{42} = 5 * 0.003 * 5 = 0.075$ → $W_{42} = 0.5 + 0.075 = 0.575$

$\Delta W_{4b} = 5 * 0.003 * 1 = 0.015$ → $W_{4b} = 0.1 + 0.015 = 0.115$

$\Delta W_{51} = 5 * 0.001 * 1 = 0.005$ → $W_{51} = 0.3 + 0.005 = 0.305$

$\Delta W_{52} = 5 * 0.001 * 5 = 0.025$ → $W_{52} = 0.6 + 0.025 = 0.625$

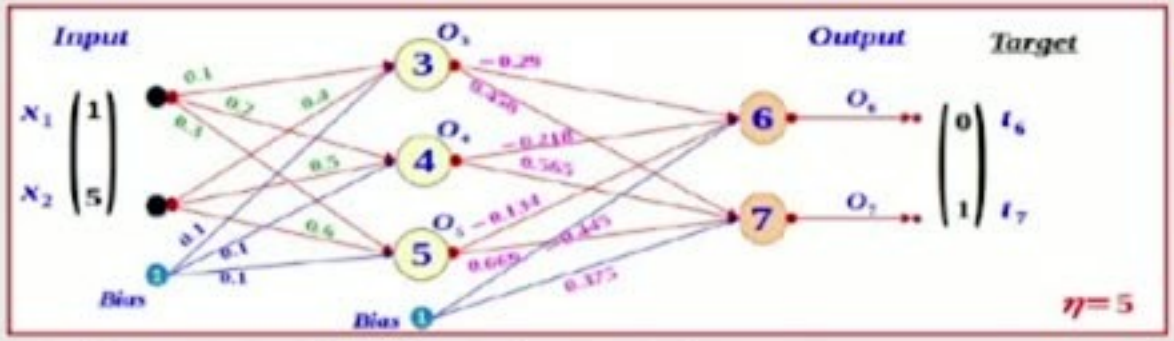
$\Delta W_{5b} = 5 * 0.001 * 1 = 0.005$ → $W_{5b} = 0.1 + 0.005 = 0.105$

# Example



$$\delta_j = O_j(1-O_j)\ [\sum_k (\delta_k w_{kj})]$$

$$\Delta_{ji} = \eta\ O_i\ O_j(1-O_j)\ [\sum_k (\delta_k w_{kj})] = \eta\ O_i\ \delta_j$$




$\delta_6 = 0.791(1-0.791)(0-0.791) = -0.131$

$\delta_7 = 0.791(1-0.791)(1-0.791) = 0.035$

$$O_3 = \frac{1}{1+e^{-2.2}} = 0.9$$

$$O_4 = \frac{1}{1+e^{-2.8}} = 0.943$$

$$O_5 = \frac{1}{1+e^{-3.4}} = 0.968$$

**Backward** ( Update ***hidden*** Layer Weights)

$\delta_3 = O_3(1-O_3)[\delta_6 * W_{63} + \delta_7 * W_{73}]$

$\delta_3 = 0.9(1-0.9)[-0.131 * -0.29 + 0.035 * 0.458] = 0.005$

$\delta_4 = 0.943(1-0.9943)[-0.131 * -0.218 + 0.035 * 0.565] = 0.003$

$\delta_5 = 0.968(1-0.968)[-0.131 * -0.134 + 0.035 * 0.669] = 0.001$

$$O_6 = \frac{1}{1+e^{-1.3312}} = 0.791$$

$$O_7 = \frac{1}{1+e^{-1.3312}} = 0.791$$

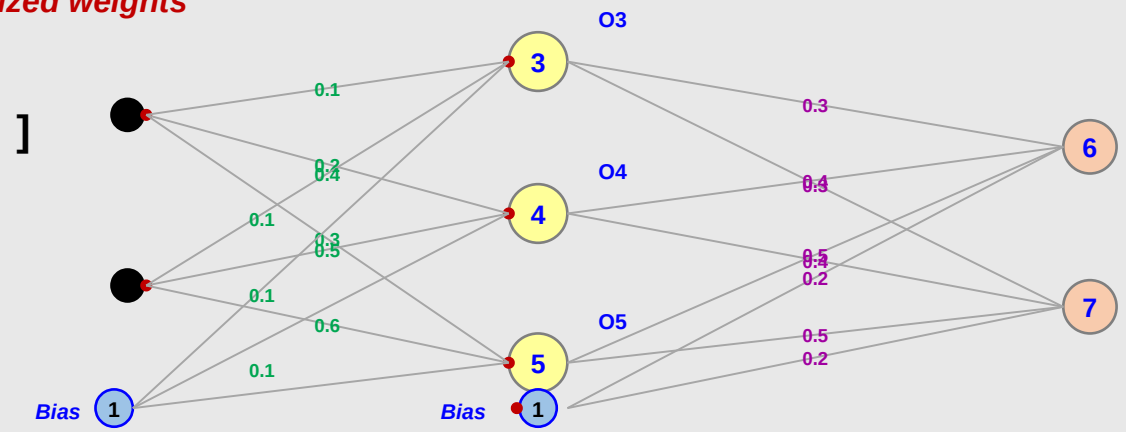
| | |
|---|---|
| $\Delta W_{31} = 5 * 0.005 * 1 = 0.025$ | $W_{31} = 0.1 + 0.025 = 0.125$ |
| $\Delta W_{32} = 5 * 0.005 * 5 = 0.125$ | $W_{32} = 0.4 + 0.125 = 0.525$ |
| $\Delta W_{3b} = 5 * 0.005 * 1 = 0.025$ | $W_{3b} = 0.1 + 0.025 = 0.125$ |
| $\Delta W_{41} = 5 * 0.003 * 1 = 0.015$ | $W_{41} = 0.2 + 0.015 = 0.215$ |
| $\Delta W_{42} = 5 * 0.003 * 5 = 0.075$ | $W_{42} = 0.5 + 0.075 = 0.575$ |
| $\Delta W_{4b} = 5 * 0.003 * 1 = 0.015$ | $W_{4b} = 0.1 + 0.015 = 0.115$ |
| $\Delta W_{51} = 5 * 0.001 * 1 = 0.005$ | $W_{51} = 0.3 + 0.005 = 0.305$ |
| $\Delta W_{52} = 5 * 0.001 * 5 = 0.025$ | $W_{52} = 0.6 + 0.025 = 0.625$ |
| $\Delta W_{5b} = 5 * 0.001 * 1 = 0.005$ | $W_{5b} = 0.1 + 0.005 = 0.105$ |

## Initialized weights

Input

x1 1

x2 5




Output

O6 0.791

O7 0.791

Target

t6 0

t7 1

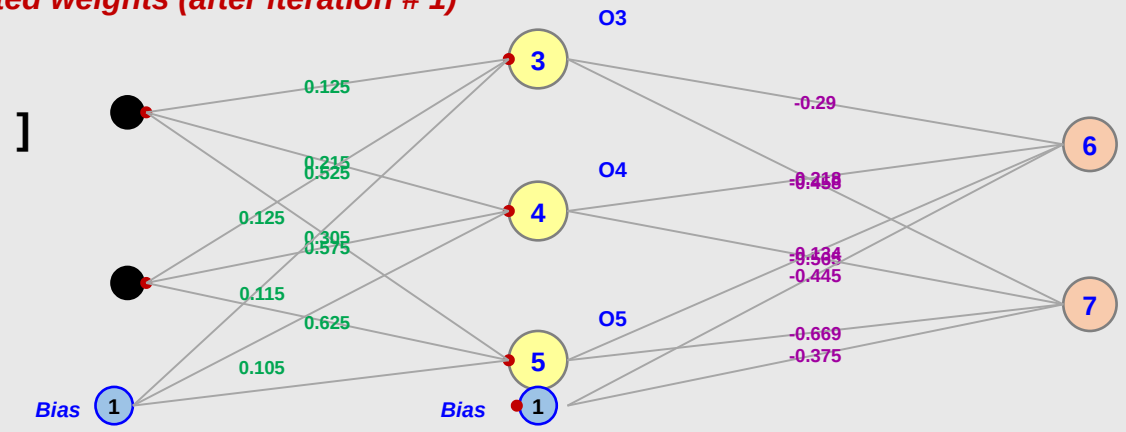
η = 5

## Updated weights (after iteration # 1)

Input

x1 1

x2 5




Output

O6 0.352

O7 0.835

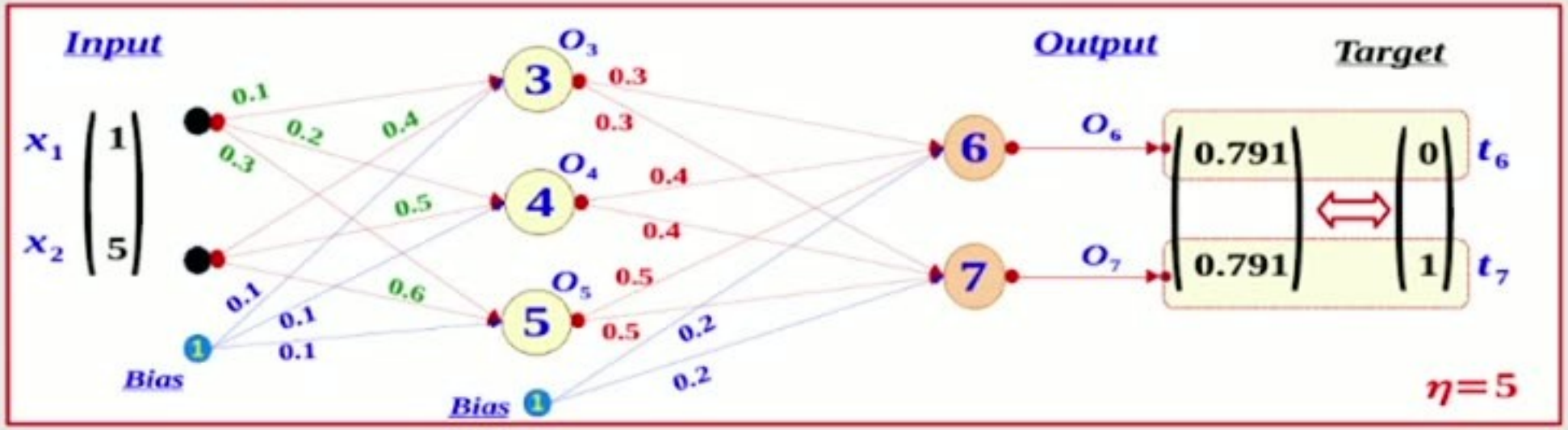
Target

t6 0

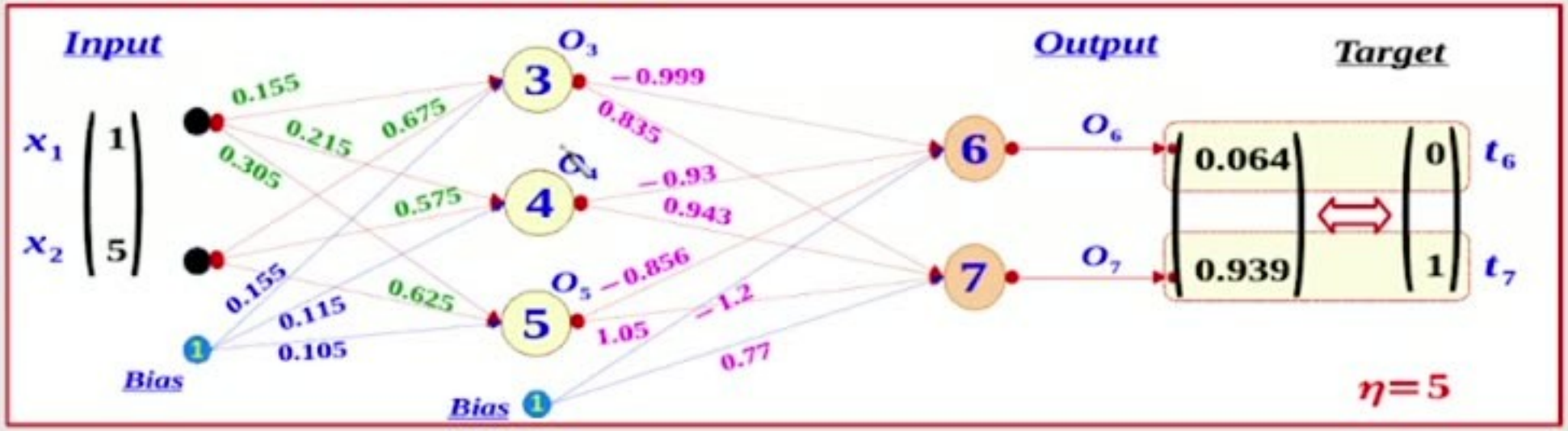
t7 1

η = 5

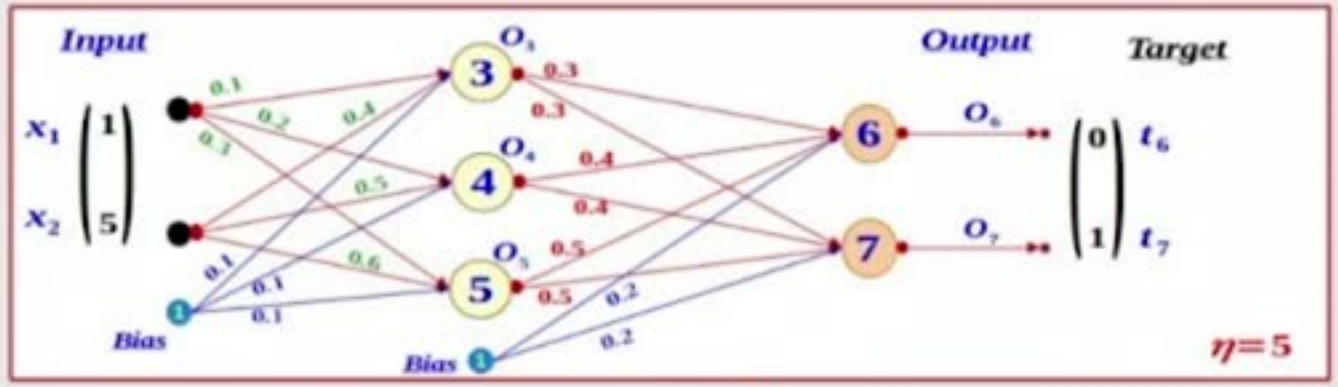
## Initialized weights




## Updated weights (after iteration # 10)



# Forward Calculations




| | $O_6$ | $O_7$ | Squared Error |
|---|---|---|---|
| Initial Weights | 0.791 | 0.791 | 0.669362 |
| Iteration 1 | 0.352 | 0.835 | 0.151129 |
| Iteration 2 | 0.152 | 0.875 | 0.038729 |
| Iteration 3 | 0.119 | 0.895 | 0.025186 |
| Iteration 4 | 0.102 | 0.908 | 0.018868 |
| Iteration 5 | 0.091 | 0.917 | 0.01517 |
| | ⋮ | ⋮ | |
| **Target** | **0** | **1** | |